

设备属性

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- 设备属性列表
- 设备属性推送
- 设备属性设置

设备属性列表

- pushMode:
 - 0: 设备推送定频数据, 设备将以 0.5HZ 的频率定时上报 (Topic:thing/product/{device_sn}/osd)
 - 1: 设备推送状态数据, 设备在状态变化时上报 (Topic:thing/product/{device_sn}/state)
- accessMode:
 - r: 属性只读
 - rw: 属性可读写 (Topic:thing/product/{gateway_sn}/property/set)

Column	Name	Type	con
home_position_is_valid	Home 点有效性	enum_int	{"0":"无效","1":"有效"}
heading	机场朝 向角	double	{"max":"180","min":"-180'°"}
rtcm_info	机场 RTK 标 定源	struct	
»mount_point	网络 RTK 挂 载点信 息	text	

»port	网络端口信息	text	
»host	网络host信息	text	
»rtcm_device_type	设备类型	enum_int	{"1":"机场"}
»source_type	标定类型	enum_int	{"0":"未标定","1":"自收敛网络RTK标定"}
wireless_link_topo	图传连接拓扑	struct	
»secret_code	加密编码	array	{"size": 28, "item_type": ir
»center_node	飞行器对频信息	struct	
»»sdr_id	扰码信息	int	
»»sn	设备sn	text	
»leaf_nodes	机场或遥控器对频信息	array	{"size": -, "item_type": stru
»»sdr_id	扰码信息	int	
»»sn	设备sn	text	
»»control_source_index	控制源序号	int	{"max":"2","min":"1","step
air_conditioner	机场空调工作状态信息	struct	

»air_conditioner_state	机场空调状态	enum_int	{ "0": "空闲模式(无制冷、制热模式)", "2": "制热模式", "3": "除雾模式", "5": "制热退出模式", "6": "制热准备模式", "8": "制热准备模式", "10": "风冷准备中", "11": "风冷准备中", "13": "除雾准备中", "14": "除雾准备中" }
»switch_time	剩余等待可切换时间	int	{ "unit_name": "秒 / s" }
air_transfer_enable	空中回传	bool	{ "false": "关闭", "true": "开启" }
silent_mode	机场静音模式	enum_int	{ "0": "非静音模式", "1": "静音模式" }
user_experience_improvement	用户体验改善计划	enum_int	{ "0": "初始状态", "1": "拒绝加入用户体验改善计划", "2": "同意加入用户体验改善计划" }
dongle_infos	4G Dongle 信息	array	{ "size": -, "item_type": struct }
»imei	dongle imei	text	
»dongle_type	Dongle 类型	enum_int	{ "6": "旧 Dongle", "10": "支持 4G 的 Dongle" }
»eid	dongle eid	text	
»esim_activate_state	eSIM 激活状态	enum_int	{ "0": "未知", "1": "未激活", "2": "已激活" }

»sim_card_state	SIM 卡状态	enum_int	{"0":"未插入","1":"已插入"}
»sim_slot	SIM 卡槽使能状态	enum_int	{"0":"未知","1":"实体 SIM"
»esim_infos	eSIM 信息	array	{"size": -, "item_type": str}
»»telecom_operator	支持的运营商	enum_int	{"0":"未知","1":"移动","2":
»»enabled	eSIM 使能状态	bool	{"false":"未使用","true":"使
»»iccid	sim iccid	text	
»sim_info	SIM 卡信息	struct	
»»telecom_operator	支持的运营商	enum_int	{"0":"未知","1":"移动","2":
»»sim_type	SIM 卡类型	enum_int	{"0":"未知","1":"其他普通
»»iccid	sim iccid	text	
drone_battery_maintenance_info	飞行器电池保养信息	struct	
»maintenance_state	保养状态	enum_int	{"0":"无需保养","1":"待保
»maintenance_time_left	电池保养剩余时间	int	{"unit_name":"小时 / h"}
»heat_state	电池加热保温状态	enum_int	{"0":"电池未开启加热或保中","2":"电池在保温中"}

»batteries	电池详细 信息	array	{ "size": -, "item_type": str
»»capacity_percent	电池剩 余电量	int	{ "max":100,"min":0}
»»index	电池序 号	enum_int	{ "0": "左电池", "1": "右电池"
»»voltage	电压	int	{ "unit_name": "毫伏 / mV"
»»temperature	温度	float	{ "unit_name": "摄氏度 / °C"
maintain_status	保养信 息	struct	
»maintain_status_array	保养信 息数组	array	{ "size": -, "item_type": str
»»state	保养状 态	enum_int	{ "0": "无保养", "1": "有保养"
»»last_maintain_type	上一次 保养类 型	enum_int	{ "0": "无保养", "17": "机场常 养" }
»»last_maintain_time	上一次 保养时 间	date	{ "unit_name": "秒 / s" }
»»last_maintain_work_sorties	上一次 保养时 作业架 次	int	{ "max": "2147483647", "mi
position_state	搜星状 态	struct	
»is_calibration	是否标 定	enum_int	{ "0": "未标定", "1": "已标定"

»is_fixed	是否收敛	enum_int	{"0":"未开始","1":"收敛中失败"}
»quality	搜星档位	enum_int	{"1":"1档","2":"2档","3":"3档","10":"RTK fixed"}
»gps_number	GPS 搜星数量	int	
»rtk_number	RTK 搜星数量	int	
emergency_stop_state	紧急停止按钮状态	enum_int	{"0":"关闭","1":"开启"}
drone_charge_state	飞行器充电状态	struct	
»capacity_percent	电量百分比	int	{"max":"100","min":"0"}
»state	充电状态	enum_int	{"0":"空闲","1":"充电中"}
backup_battery	机场备用电池信息	struct	
»switch	备用电池开关	enum_int	{"0":"关闭","1":"开启"}
»voltage	备用电池电压	int	{"desc":"备用电池关闭时E0","max":"30000","min":"(伏 / mV)"}
»temperature	备用电池温度	float	{"step":"0.1","unit_name":
alarm_state	机场声光报警状态	enum_int	{"0":"关闭","1":"开启"}
battery_store_mode	电池运行模式	enum_int	{"1":"计划模式","2":"待命

activation_time	机场激活时间 (unix 时间戳)	int	{"unit_name": "秒 / s"}
height	椭圆高度	double	{"unit_name": "米 / m"}
alternate_land_point	备降点	struct	
»longitude	经度	float	{}
»latitude	纬度	float	{}
»safe_land_height	安全高度(备降转移高)	float	{}
»is_configured	是否设置备降点	enum_int	{"0": "未设置", "1": "已设置"}
»height	椭圆高度	float	
compatible_status	固件一致性	enum_int	{"0": "不需要一致性升级",
acc_time	机场累计运行时长	int	{"unit_name": "秒 / s"}
first_power_on	首次上电时间	int	{"unit_name": "毫秒 / ms"}

storage	存储容量	struct	
»total	总容量	int	{"unit_name":"千字节 / KI
»used	已使用容量	int	{"unit_name":"千字节 / KI
working_current	工作电流	float	{"unit_name":"毫安 / mA"
working_voltage	工作电压	int	{"unit_name":"毫伏 / mV"
humidity	舱内湿度	float	{"max":"100","min":"0","st 湿度 / %RH"}
temperature	舱内温度	float	{"unit_name":"摄氏度 / °C
environment_temperature	环境温度	float	{"unit_name":"摄氏度 / °C
wind_speed	风速	float	{"unit_name":"米每秒 / m
rainfall	降雨量	enum_int	{"0":"无雨","1":"小雨","2":
live_capacity	网关设备直播能力	struct	
»available_video_number	可选择推流的码流数量	int	
»coexist_video_number_max	可同时推流的最大码流数量	int	
»device_list	可选择的视频设备源	array	{"size": -, "item_type": str
»»sn	飞行器 等视频	text	

	序列号 (SN)		
»»available_video_number	该序列号设备可以被选择推流的码流数	int	
»»coexist_video_number_max	该序列号设备可以同时被推流的码流数	int	
»»camera_list	该序列号设备上的相机列表	array	{"size": -, "item_type": str
»»»camera_index	相机索引	text	
»»»»available_video_number	该相机级别的视频源可以被选择推流的码流数	int	
»»»»coexist_video_number_max	该相机级别的视频源可以同时被推流的码流数	int	

»»»»video_list	该相机级别的视频源可以选择的码流列表	array	{"size": -, "item_type": str
»»»»»video_index	该相机级别的视频源可以选择的码流索引	text	
»»»»»video_type	该相机级别的视频源可以选择的码流类型	text	
»»»»»switchable_video_types	该视频流支持切换的视频镜头类型	array	{"size": -, "item_type": tex
live_status	网关当前整体直播状态推送	array	{"size": -, "item_type": str
»video_id	直播码流标识符	text	
»video_type	视频类型	text	{"length": "24"}
»video_quality	直播码流的质量	enum_int	{"0": "自适应", "1": "流畅", "2": "清晰"}

»status	直播状态	enum_int	{"0":"未直播","1":"在直播"}
»error_status	错误码	int	{"length":6}
wireless_link	图传链路	struct	
»dongle_number	飞行器上 Dongle 数量	int	
»4g_link_state	4G 链路连接状态	enum_int	{"0":"断开","1":"连接"}
»sdr_link_state	SDR 链路连接状态	enum_int	{"0":"断开","1":"连接"}
»link_workmode	机场的图传链路模式	enum_int	{"0":"SDR 模式","1":"4G 模式"}
»sdr_quality	SDR 信号质量	int	{"max":"5","min":"0","step":1}
»4g_quality	总体 4G 信号质量	int	{"max":"5","min":"0","step":1}
»4g_uav_quality	天端 4G 信号质量	int	{"max":"5","min":"0","step":1}
»4g_gnd_quality	地端 4G 信号质量	int	{"max":"5","min":"0","step":1}
»sdr_freq_band	SDR 频段	float	
»4g_freq_band	4G 频段	float	

media_file_detail	媒体文件上传细节	struct	
»remain_upload	待上传数量	int	
job_number	机场累计作业次数	int	{"unit_name":"次 / count"}
drone_in_dock	飞行器是否在舱	enum_int	{"0":"舱外","1":"舱内"}
network_state	网络状态	struct	
»type	网络类型	enum_int	{"1":"4G","2":"以太网"}
»quality	网络质量	enum_int	{"0":"无信号","1":"差","2":"好","5":"好"}
»rate	网络速率	float	{"unit_name":"千字节每秒"}
supplement_light_state	补光灯状态	enum_int	{"0":"关闭","1":"打开"}
cover_state	舱盖状态	enum_int	{"0":"关闭","1":"打开","2":
sub_device	子设备状态	struct	
»device_sn	子设备序列号 (SN)	text	
»device_model_key	子设备枚举值	text	
»device_online_status	机场停机坪上的飞行	enum_int	{"0":"关机","1":"开机"}

	属性名称	数据类型	取值范围
»device_paired	机场停机坪上的飞行器是否与机场对频	enum_int	{ "0": "未对频", "1": "已对频" }
flighttask_step_code	机场任务状态	enum_int	{ "0": "作业准备中", "1": "飞行中", "2": "飞行恢复", "3": "自定义飞行区域中", "4": "任务空闲", "5": "任务结束", "255": "未知状态" }
mode_code	机场状态	enum_int	{ "0": "空闲中", "1": "现场调试", "2": "升级中", "3": "任务中", "4": "任务空闲", "5": "任务结束" }
firmware_upgrade_status	固件升级状态	enum_int	{ "0": "未升级", "1": "升级中", "2": "升级失败" }
firmware_version	固件版本	text	{ "length": "64" }
latitude	纬度	double	{ "max": "90", "min": "-90", "step": "0.000001" }
longitude	经度	double	{ "max": "180", "min": "-180", "step": "0.000001" }
drc_state	DRC 链路的状态	enum_int	{ "0": "未连接", "1": "连接中", "2": "连接失败" }

设备属性推送

Topic: thing/product/{device_sn}/state 状态数据:设备在状态变化时上报

Topic: thing/product/{device_sn}/osd 定频数据:设备将以 0.5HZ 的频率定时上报

Direction: up

API 说明:

负载属性上报是指飞行器挂载的负载的属性上报，如相机的属性上报。一个负载由负载索引唯一确定（负载索引：产品类型-子类型-挂载位置 {type-subtype-gimbalIndex}），type、subtype。数值请参考：产品支持[🔗](#)，gimbalindex的对应关系请查看 [航线文件格式](#)[🔗](#)中的 wpml:payloadPositionIndex。

横滚的角度等。对于 [直播能力更新](#) ，负载属性上报的内容更多为相机在直播时的能力，如最大可同时推流的码流数量。负载属性具体结构体请参考下文的example。需要注意遥控器的设备属性在一条消息体中上报，而机场的设备属性推送是分多条推送的。

Data:

Column	Name	Type	constraint	Description
data	消息内容	text	{}	内容可参考设备的设备属性
»camera_index	type-subtype-gimbalIndex	text	{}	gimbalIndex为相机位置，type、sub_type。参考： 产品支持

Example:

Topic: thing/product/{dock_sn}/state

Topic: thing/product/{dock_sn}/osd

```
{
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1667220873846,
  "data": {
    "job_number": 492,
    "acc_time": 1859010,
    "activation_time": 0,
    "maintain_status": {
      "maintain_status_array": [
        {
          "state": 0,
          "last_maintain_type": 17,
          "last_maintain_time": 0,
          "last_maintain_work_sorties": 0
        }
      ]
    },
    "working_voltage": 25440,
    "working_current": 1120,
    "backup_battery": {
      "voltage": 26631,
      "temperature": 27.9,
      "switch": 1
    }
  },
}
```

```

    }
  },
  "gateway": "dock_sn"
}

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "flighttask_step_code": 255,
    "media_file_detail": {
      "remain_upload": 0
    },
    "wireless_link": {
      "4g_freq_band": 2.4,
      "4g_gnd_quality": 0,
      "4g_link_state": 0,
      "4g_quality": 0,
      "4g_uav_quality": 0,
      "dongle_number": 0,
      "link_workmode": 0,
      "sdr_freq_band": 2.4,
      "sdr_link_state": 0,
      "sdr_quality": 0
    }
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1667220881576,
  "gateway": "dock_sn"
}

```

```

{
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "timestamp": 1667220916697,
  "data": {
    "network_state": {
      "type": 2,
      "quality": 0,
      "rate": 5.0970001220703125
    },
    "drone_charge_state": {
      "state": 0,
      "capacity_percent": 100
    },
    "drone_in_dock": 1,
    "rainfall": 0,
    "wind_speed": 0,
    "environment_temperature": 24,
  }
}

```

```

    "longitude": 113.703482143,
    "height": 34.174125671386719,
    "alternate_land_point": {
      "latitude": 22.907898319908661,
      "longitude": 113.70347329676635,
      "safe_land_height": 0,
      "is_configured": 1
    },
    "first_power_on": 1631945855969,
    "position_state": {
      "is_calibration": 1,
      "is_fixed": 2,
      "quality": 5,
      "gps_number": 6,
      "rtk_number": 25
    },
    "storage": {
      "total": 82045336,
      "used": 51772
    },
    "mode_code": 1,
    "cover_state": 0,
    "supplement_light_state": 0,
    "emergency_stop_state": 0,
    "air_conditioner": {
      "air_conditioner_state": 3,
      "switch_time": 1
    },
    "battery_store_mode": 1,
    "alarm_state": 0,
    "putter_state": 0,
    "sub_device": {
      "device_sn": "1581F5BKD225D00BP891",
      "device_model_key": "0-67-0",
      "device_online_status": 0,
      "device_paired": 1
    }
  },
  "gateway": "dock_sn"
}

```

Topic: thing/product/{*aircraft_sn*}/state

Topic: thing/product/{*aircraft_sn*}/osd

```

{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx",
  "data": {
    "52-0-0": {

```



```

        "measure_target_latitude": 0,
        "measure_target_longitude": 0,
        "payload_index": "52-0-0",
        "version": 1
    },
    "activation_time": 1667935211,
    "attitude_head": 41.7,
    "attitude_pitch": 2.7,
    "attitude_roll": 0,
    "battery": {
        "batteries": [
            {
                "capacity_percent": 95,
                "firmware_version": "02.00.20.44",
                "high_voltage_storage_days": 16,
                "index": 0,
                "loop_times": 137,
                "sn": "4BUPJ99DAD009W",
                "sub_type": 0,
                "temperature": 33.3,
                "type": 0,
                "voltage": 24303
            },
            {
                "capacity_percent": 85,
                "firmware_version": "02.00.20.44",
                "high_voltage_storage_days": 5,
                "index": 1,
                "loop_times": 82,
                "sn": "4BUPJ9EDAD01CE",
                "sub_type": 0,
                "temperature": 32,
                "type": 0,
                "voltage": 24311
            }
        ],
        "capacity_percent": 90,
        "landing_power": 0,
        "remain_flight_time": 0,
        "return_home_power": 0
    },
    "distance_limit_status": {
        "distance_limit": 5000,
        "state": 0
    },
    "elevation": 0,
    "firmware_version": "05.01.0214",
    "gear": 1,
    "height": 38.417465209960938,
    "height_limit": 120,
    "home_distance": 0,
    "horizontal_speed": 0,

```

```

        "maintain_status_array": [
            {
                "last_maintain_flight_time": 0,
                "last_maintain_time": 0,
                "last_maintain_type": 1,
                "state": 0
            },
            {
                "last_maintain_flight_time": 0,
                "last_maintain_time": 0,
                "last_maintain_type": 2,
                "state": 0
            },
            {
                "last_maintain_flight_time": 0,
                "last_maintain_time": 0,
                "last_maintain_type": 3,
                "state": 0
            }
        ]
    },
    "mode_code": 0,
    "night_lights_state": 0,
    "obstacle_avoidance": {
        "downside": 1,
        "horizon": 1,
        "upside": 1
    },
    "position_state": {
        "gps_number": 0,
        "is_fixed": 0,
        "quality": 0,
        "rtk_number": 0
    },
    "storage": {
        "total": 0,
        "used": 0
    },
    "total_flight_distance": 0,
    "total_flight_sorties": 0,
    "total_flight_time": 0,
    "track_id": "",
    "vertical_speed": 0,
    "wind_direction": 0,
    "wind_speed": 0
},
"tid": "2d2040eb-23b0-43dc-b7ac-64838276c4ac",
"timestamp": 1670422793916,
"gateway": "dock_sn"
}

```

设备属性设置

Topic: thing/product/{gateway_sn}/property/set

Direction: down

Data:

Column	Name	Type	constraint	Description
data	消息内容	text	{}	内容可参考设备的设备属性

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx",
  "data": {
    "distance_limit_status": {"state": 1}
  },
  "tid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx",
  "timestamp": 1643268212187
}
```

Topic: thing/product/{gateway_sn}/property/set_reply

Direction: up

Data:

Column	Name	Type	constraint	Description
data	消息内容	text	{}	内容可参考设备的设备属性

Example:

```
{
  "bid": "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxx",
  "data": {
    "distance_limit_status": {
      "state": {
        "result": 0 // 0: 成功, 1: 失败, 2: 超时, 0x123456 具体的错误
      }
    }
  }
}
```

}

设备管理

请问您的义怕阅读体验如何？